

11.3

$$(6) \quad \left| \frac{1}{x^b} \right| \leq \frac{1}{x^b}$$

而 $b \in (0, 1)$, 则 $\int_0^1 \frac{1}{x^b} dx$ 收敛.

故 $\int_0^1 \frac{dx}{x^b}$ 一致收敛.

$$(2) \quad \frac{x^a - x^b}{\ln x} = \int_b^a x^y dy, \quad \text{则 } \Gamma(x) = \int_0^1 dx \int_b^a x^y dy$$

而 $\int_0^1 x^y dx$ 为常积分, 一致收敛.

$$\therefore \Gamma(x) = \int_b^a dy \int_0^1 x^y dx = \int_b^a \frac{x^{y+1}}{y+1} \Big|_0^1 dy = \int_b^a \frac{1}{y+1} dy = \ln \frac{a+1}{b+1}$$

4,

$$(1) \quad \Gamma(x) = \int_0^1 x^x \sqrt{1-x} dx = B\left(\frac{4}{3}, \frac{4}{3}\right) = \frac{\Gamma\left(\frac{4}{3}\right) \Gamma\left(\frac{4}{3}\right)}{\Gamma\left(\frac{8}{3}\right)}$$

$$(5) \quad \text{令 } t = x^4, \quad \frac{1}{4} \int_0^1 t^{-\frac{3}{4}} (1-t)^{-\frac{1}{2}} dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4} \cdot \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

$$(9) \quad \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx = \frac{1}{2} B\left(\frac{n+1}{2}, \frac{m+1}{2}\right)$$

12.1

$$1. \quad \int_{-\pi}^{\pi} \sin mx \cdot \sin nx dx = \frac{1}{2} \int_{-\pi}^{\pi} (\cos(m-n)x - \cos(m+n)x) dx$$

$$= -\frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} - \frac{\sin(m-n)x}{m-n} \right] \Big|_{-\pi}^{\pi} = 0$$

$$\int_{-\pi}^{\pi} \sin mx \cos nx dx = \frac{1}{2} \int_{-\pi}^{\pi} (\sin(m+n)x + \sin(m-n)x) dx$$

$$= -\frac{1}{2} \left[\frac{\cos(m+n)x}{m+n} + \frac{\cos(m-n)x}{m-n} \right] \Big|_{-\pi}^{\pi} = 0$$

$$3. \|f - \lambda g\|^2 \geq 0$$

$$\frac{1}{2} \int_{-2}^2 (f - \lambda g)^2 dx \geq 0$$

$$\frac{\lambda^2}{2} \int_{-2}^2 g^2 dx - \frac{2\lambda}{2} \int_{-2}^2 f \cdot g dx + \frac{1}{2} \int_{-2}^2 f^2 dx \geq 0$$

$$\lambda^2 \|g\|^2 - 2\lambda (f, g) + \|f\|^2 \geq 0$$

$$\therefore \Delta = 4(f, g)^2 - 4\|g\|^2\|f\|^2 \leq 0$$

$$\therefore |(f, g)| \leq \|g\| \cdot \|f\|$$

12.2.

$$1. (3) f(x) = |x|$$

易知 $f(x)$ 为偶函数

$$a_0 = \frac{2}{2} \int_0^2 x \cdot dx = 2$$

$$a_n = \frac{2}{2} \int_0^2 x \cdot \cos nx \cdot dx = \frac{2}{2} \cdot \frac{1}{n} \int_0^2 x \cdot d \sin nx = \frac{2}{2n} \left[x \sin nx \Big|_0^2 - \int_0^2 \sin nx dx \right]$$

$$= \frac{2}{2n^2} (\cos nx \Big|_0^2)$$

$$= \frac{2}{2n^2} [(-1)^n - 1]$$

$$\therefore f(x) \sim \frac{2}{2} + \sum_{n=1}^{\infty} \frac{2}{2n^2} [(-1)^n - 1] \cos nx$$

$$= \frac{2}{2} + \sum_{n=1}^{\infty} \frac{-4}{2(2n-1)^2} \cos(2n-1)x$$

$$= \frac{2}{2} - \frac{4}{2} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}$$

$\therefore f(x)$ 为连续且单调, 不存在间断点

$\therefore f(x)$ 在连续点处收敛于傅立叶级数

$$\therefore S(x) = f(x), x \in (-2, 2)$$

$$(5) f(x) = \sin^4 x, -2 \leq x \leq 2$$

$f(x)$ 为偶函数

$$a_0 = \frac{2}{2} \int_0^2 \sin^4 x dx = \frac{4}{2} \cdot \frac{2}{2} \cdot \frac{3!!}{4!!} = \frac{3}{4}$$

$$a_n = \frac{2}{2} \int_0^2 \sin^4 x \cdot \cos nx \cdot dx = \frac{2}{2} \int_0^2 \frac{1 - 2\cos 2x + \cos^2 2x}{4} \cdot \cos nx \cdot dx$$

$$= \frac{2}{2} \int_0^2 \left(-\frac{1}{2} \cos 2x \cos nx + \frac{1}{8} \cos 4x \cos nx \right) dx$$

$$\text{① 当 } n=2 \text{ 时, } a_2 = -\frac{1}{2} \int_0^2 \cos^2 2x dx = -\frac{1}{2}$$

$$\text{当 } n=4 \text{ 时, } a_4 = \frac{1}{4} \cdot \frac{1}{2} \cdot \int_0^{2\pi} \cos^2 4x \cdot dx = \frac{1}{8}$$

$$n \neq 2/4 \text{ 时, } a_n = 0$$

$$\therefore f(x) \sim \frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x$$

又有 $f(x)$ 分片连续且单调, 无间断点

$$S(x) = f(x), x \in (-2, 2)$$

5. $f(x) = 3$ 是奇函数

$$a_n = 0$$

$$\begin{aligned} b_n &= \frac{2}{2\pi} \int_0^{2\pi} f(x) \sin nx \, dx = \frac{6}{2\pi} \int_0^{2\pi} \sin nx \, dx \\ &= \frac{-6}{n2\pi} (\cos nx \Big|_0^{2\pi}) \\ &= \frac{-6}{n2\pi} [(-1)^n - 1] \\ &= \frac{6}{n2\pi} [(-1)^{n+1} + 1] \end{aligned}$$

$$\begin{aligned} \therefore f(x) &\sim \sum_{n=1}^{\infty} \frac{6}{n2\pi} [(-1)^{n+1} + 1] \sin nx \\ &= \sum_{n=1}^{\infty} \frac{12}{(2n-1)2\pi} \sin (2n-1)x \\ &= \frac{12}{2\pi} \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1}, \quad 0 < x < 2\pi \end{aligned}$$

而 $f(x)$ 分片连续单调, 故收敛于傅里叶级数

$$\begin{aligned} \therefore \frac{12}{2\pi} \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1} &= 3 \\ \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1} &= \frac{2\pi}{4} \end{aligned}$$

取 $x = \frac{\pi}{2}$, 即有

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{2\pi}{4}$$